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Underwater Loop-Closure Detection for Mechanical Scanning Imaging Sonar by Filtering the Similarity Matrix with Probability Hypothesis Density Filter

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ABSTRACT Robust and accurate estimation of position and attitude of a UUV (Unmanned Underwater Vehicle) from sonar scans is essential for simultaneous localization and mapping (SLAM). Both dead-reckoning based on the inertial navigation system and the motion parameter estimation based on the registration of the ultrasound scan sequence can contribute to the performance of the system. However, the rapidly-growing accumulated error tends to counteract the precise localization of the vehicle. In this paper, a method for loop-closure detection is proposed that adjusts the accumulated error for the underwater acoustic SLAM when the vehicle scans the underwater environment using an Mechanical Scanning Imaging Sonar (MSIS). Firstly, a similarity matrix for pairs of scans is constructed to represent the loop-closing tracks. In the registration step, two novel features, namely the intensity projection histograms and a polar gradient matrix, are extracted to calculate the translational and rotational parameters respectively. Secondly, the probability hypothesis density (PHD) filter is used to extract the possible loop-closure constraints from the similarity matrix, removing the random noise brought by accidental correlation and refining the concurrent loop-closing tracks resulted from long-range scanning. Lastly, the loop-closure constraints from the refined loop-closing tracks are fed into the GraphSLAM system to adjust the pose of each scan by constraint optimization. Experiments on the MSIS sonar images collected in structured and unstructured underwater environments validate the effectiveness of the proposed loop-closure detection method.

INDEX TERMS Forward-looking sonar, intensity projection histogram, PHD filter, polar gradient matrix, underwater loop-closure detection.

I. INTRODUCTION

When a vehicle observes its underwater environment with a high-frequency forward-looking sonar at short distances, it is an important task to transform the recorded sonar image sequence into a globally consistent map which is known as simultaneous localization and mapping (SLAM) [1], [2]. Not only does a panoramic map facilitate the investigation and interpretation of an area of interest, such as underwater mine countermeasures [3], and underwater chain inspection [4], but it also helps UUVs (Unmanned Underwater

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Vehicles) to navigate and act autonomously which includes self-localization [5], self-planning [6], and autonomous intervention [7].

The accuracy of the global map depends essentially on the localization precision of vehicle. Due to the rapid attenuation of electromagnetic waves in water, GPS (Global Positioning System) signals are inaccessible to underwater robots. Therefore, inertial navigation systems are used to estimate the position of the robot. Unfortunately, the drift error inherent in the dead-reckoning process grows with time.

One solution to reduce the accumulated global error is an acoustic localization system, such as the long base line (LBL) [8] or ultra-short base line (USBL) [9] techniques, that updates the absolute position of the vehicle periodically based on the reference information from floats, subsurface buoys, or a nearby mother ship. Unfortunately, the auxiliary facilities and devices are unavailable in many situations.

A more effective strategy is to derive reference information from exteroceptive sensors. For example, when a robot senses a target that it has seen before, then the target can be regarded as a reference, and the dead-reckoning error between the two visits can be adjusted accordingly. This revisit detection is widely known as loop-closure detection which, together with image registration (or dead reckoning when available), constitute the two pillars of the sensory data processing in a SLAM system. Benefiting from the loop-closure detection, the localization of the vehicle and the construction of the map are solved simultaneously in the SLAM system, reducing the position estimation error while generating a consistent map simultaneously. It has been suggested in Ref. [10] that visual-appearance-based approaches are capable of detecting loop closures even when the vehicle locations provided by the dead-reckoning mechanism do not contain usable information.

Optical camera and sonar are two common sensors for underwater vehicles. Although optical cameras have a much higher resolution, light-based sensing suffers a shorter scanning distance in turbid water. Therefore, sonars that are much less sensitive to the water quality and that have a longer visual distance, are widely used sensors in underwater environments. In this paper, we aim at building an underwater SLAM system with the mechanical scanning imaging sonar (MSIS), with the emphasis on acoustic loop-closure detection.

As a forward-looking sonar, the MSIS scans the environment step by step. A transducer emits a beam in a certain direction and, if an object is found in this direction, then the beam will be reflected such that the information about the object can reach the transducer. After that, MSIS emits another beam along the next predetermined direction and obtains the corresponding response. The process is repeated until the full scanning sector is covered. In this way, MSIS images have a long scanning period, leading to coarse temporal resolution. Moreover, due to the long wavelength of acoustic waves and the scattering of sound waves by water, sonar images have not only a lower spatial resolution, but also contain strong speckle noise. Low temporal and spatial resolution severely challenges traditional image processing methods, like feature extraction, image registration as well as loop-closure detection.

In this paper, a method for loop-closure detection is proposed that is based on filtering of the similarity matrix and which has the capability to correct the globally positional error accumulated in the dead-reckoning process. The method draws upon two particular features, namely the intensity projection histograms and the polar gradient matrix, which are capable of extracting the energy distribution along two perpendicular projection axes and the azimuthal direction for each sonar scan. Based on feature correlation, the sonar scans are coarsely registered to construct a similarity matrix whose elements measure the similarity score between all possible pairs of current and previous sonar scans. The similarity matrix is mainly composed of true or apparent loop-closing track segments which are due to strong overlaps between sonar scans collected during two subsequent visits of the same or of two similar locations. Then, we analogously take the loop-closing tracks as the trajectories of multiple virtual targets, and use a probability hypothesis density (PHD) filter [11] to extract the actual loop-closure constraints from the similarity matrix. Finally, the loop-closure constraints are fed into the GraphSLAM system to adjust the position and pose of each scan and generate a panoramic map. Two MSIS datasets were taken from real underwater environments: one from a man-made structured environment and the other from a natural environment. They are used to test the feasibility and efficiency of the proposed loop-closure detection method.

Our main contributions are summarized as follows:

- We present a data-processing pipeline that solely depends on data associations between sonar scans to generate loop-closure candidates, independently of pose estimation from dead reckoning, for the underwater acoustic SLAM system.
- Two novel features, the intensity projection histogram and the polar gradient matrix, are designed to describe the low resolution sonar scans, with which the translational and rotational parameters can be estimated and the similarity matrix is constructed.
- The loop-closing tracks in the similarity matrix are analogously generated by multiple virtual targets and a PHD filter is adopted to extract multiple loop-closing constraints.

The remainder of the paper is organized as follows: related works are reviewed in section II. Section III gives the overview of the proposed method whose main modules, including similarity matrix construction, loop-closure detection and global error adjustment are explained through sections IV to VI. Experimental results are reported in section VII, and the paper is concluded in section VIII.

II. RELATED WORKS

Loop-closure detection corrects the accumulated global error by sensor data association. Thus, the data association method used in loop-closure detection is directly determined by the

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data form that a SLAM system relies on when building maps [12]. Generally, maps can be categorised according to the type of information used in the generation of the map.

Feature-based maps are composed of a collection of predefined geometric primitives in specific locations. The feature sets are typically extracted from the raw sensor data for feature association. For example, Ribas *et al.* [2] extracted line feature from the MSIS scans that were collected in a man-made marina environment. Williams *et al.* [13] and He *et al.* [14] took image pixel blobs as local features for feature matching. Any ambiguous features are distinguishable only by their global or relative locations. Therefore, data association methods for loop-closure detection, such as gated nearest-neighbor [15] or joint compatibility branch-and-bound (JCBB) [16] methods, depend on accurate pose estimates. Additionally, because of the low resolution, it is hard to extract discriminative features from sonar scans, especially in unstructured underwater environments.

Location-based maps discretize the environment into a number of regions, to each of which a value indicating the probability or degree of belief of occupancy is assigned. A well-known example is the occupancy grid map, which has been applied in several marine tasks. In mapping a flooded sinkhole, Farfield *et al.* [17] designed a 3D evidence grid to represent the 3D underwater tunnels, where an array of 54 pencil-beam sonars installed in the shape of three great circles provided a constellation of range measurements around the vehicle. In their effort to explore and map ancient cisterns, White *et al.* [18] also used the occupancy grid to represent the belief state of the environment. Although specific loop-closure detection methods were not mentioned in Ref. [17], [18], a location-based data association will help in reducing the error in the dead-reckoning system.

View-based maps are composed of a collection of full sensor readings, including a sequence of scans and their corresponding pose information. It represents the environment directly with the preliminary sensor data without feature extraction and matching, or data fusion. In the loop-closure detection step, Mallios *et al.* [1], [19] proposed that when a scan obtained by vehicle, it needs to be registered with all the scans whose corresponding poses fall into the neighboring range of the pose of the current scan. If the number of corresponding sonar points exceeds a certain threshold, a loop-closure constraint is added to the two poses. Furthermore, Chen *et al.* [20] used the Mahalanobis distance to calculate the range between historical poses and current vehicle pose. Note that Ma *et al.* [21] and Palomer *et al.* [22] directly compared current view to all historical views for loop-closure detection, without using dead-reckoning data.

Measuring the similarity between each sonar scan pair may cause high uncertainty in closed-loop detection. Instead, evaluating the similarity between scan sequences or local scenes can lead to greater stability. Ho *et al.* [23] pointed out that extracting sequences of similar scenes from the similarity matrix supported more reliable loop-closure detection despite the presence of repetitive and visually ambiguous scenes. In SeqSLAM proposed by Milford *et al.* [24], it was shown that the place could be precisely recognized by image sequences matching even in a visually challenging environment. In Ref. [25], image sequence is divided into groups at fixed time intervals, and the group that has the highest overall similarity with the current image is considered to be the matching sequence. In Ref. [26], an individual-visual-word-vector (I-VWV) was extracted from each image, and their combination, named sequence-visual-word-vector (S-VWV), was proposed to describe the physical scene. Then, S-VWV matching firstly detected the supposedly revisited place, and loop-closing frames were finally determined by I-VWV matching.

From the above discussion, we conclude that intuitive loop-closure detection methods, which rely on measuring the similarity between current scan and all historical scans, may generate false positive loop-closure constraints, due to the correlation between similar scenes in the natural underwater environment. On the contrary, the methods based on image sequence matching are potential to get more stable loop-closure detection results. In this paper, we also propose a method based on scan sequence matching to detect loop closures by filtering the similarity matrix with PHD filter. The proposed loop-closure detection pipeline and the experimental results are presented in details in the remainder of the paper.

III. OVERVIEW OF THE PROPOSED METHOD

In this paper, we propose an underwater loop-closure detection method that extracts the loop-closure constraints from the similarity matrix of sonar scans with a PHD filter. The framework, displayed in Fig. 1, is composed of three main steps: construction of the similarity matrix, detection of loop closure, and adjustment of the global error.

Similarity matrix construction: Each element in the similarity matrix is obtained by coarsely registering the two corresponding sonar images. In estimating the rotational parameter, the sonar image is divided into a set of grid cells in polar coordinates. For any two grid cells in each sector, we encode the intensity gradient and curvature as a binary tuple. Then, the tuples in each sector are concatenated into a binary column, and a polar gradient matrix is thus obtained. The rotational parameter could be solved by minimizing the cross-correlation between the binary matrixes. In estimating the translational parameters, we project the sonar scans into two intensity projection histograms along two specifically selected orthogonal axes. Similarly, the translational parameters can be determined by locating the maximum in the cross-correlation function between two intensity projection histograms. Finally, the similarity score between the floating scan and reference scan is obtained by calculating the overlap between the reference scan and the transformed floating scan when the rotational and translational parameters are applied.

Loop-closure detection: The scanning range of the MSIS can be up to tens of meters, a scan will be correlated with several historical scan segments. Therefore, multiple loop-

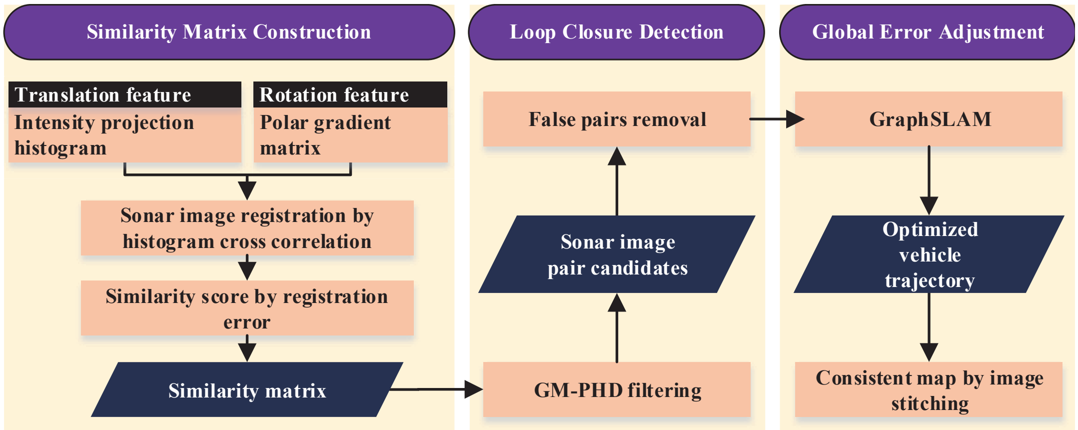

FIGURE 1. Framework of the proposed loop-closure detection method.

closure tracks may be simultaneously rendered in the similarity matrix. Taking the number of those tracks varying randomly into consideration, PHD filter is adopted to extract those tracks. It has been demonstrated in Ref. [11] that the PHD filter is appropriate to tackle tracking problems where the number of targets is not known.

Global error adjustment: The loop-closure constraints are fed into the GraphSLAM system to adjust the position and pose of each scan by constrained optimization. The error between the measurement from sonar scan registration and the estimation from dead-reckoning system is minimized to correct the accumulated error in location and attitude estimation of the vehicle.

IV. SIMILARITY MATRIX CONSTRUCTION

Denote the similarity matrix as $M = \{m_{ij}\}_{i,j=1,\ldots,N}$, where N is the number of scans. The similarity score m_{ij} of two sonar scans, i.e., floating scan I_i and reference scan I_j, is defined by the overlap after scan registration,

$$m_{ij} = g\left(f\left(I_i, \Psi\right), I_j\right). \tag{1}$$

The parameter set $\Psi = \{\Delta x^*, \Delta y^*, \Delta\theta^*\}$ transforms the floating scan to the coordinate of the reference scan, with Δx^*, Δy^* being the translational parameters, $\Delta\theta^*$ being the rotational parameter. To alleviate the noise caused by the accidental correlation between acoustic scans, only the scores that are smaller than a given threshold are used in further processing.

It is difficult to calculate the transformation parameters Ψ by the traditional feature point projection [27], [28], because discriminative feature points are hardly extractable from the MSIS scans. Two novel features, namely an intensity projection histogram and a polar gradient matrix, and their corresponding correlation function, are designed in the following subsections to estimate the rotational and translational parameters.

Note that for a long scanning period, the movements of the robot will seriously distort the sonar scans, causing large estimation errors in the subsequent feature correlation procedure. Fortunately, it can be always assumed that the local dead-reckoning error in the navigation system is small if there is no U-turn rotation. Therefore, all the scan lines in the same scanning period should be projected to the same coordinate with the location and pose information from the dead-reckoning system. For more information about this principle for correction, please refer to [19].

A. POLAR GRADIENT MATRIX AND ROTATIONAL PARAMETER ESTIMATION

We will now design a feature descriptor to model the energy distribution along the azimuthal angle, named polar gradient matrix, which will be further used to estimate the rotational parameter.

1) Rotation Feature Matrix

To model the intensity distribution along the azimuthal angle and the imaging distance, we divide the sonar image into grid cells in the polar coordinate (Fig. 2(a)). The unit angle $d\theta$ and unit distance dD should compromise with the computation cost and the estimation precision. Then, the number of angular bins and the distance bins are $N_a = \frac{360}{d\theta}$ and $N_d = \frac{D}{dD}$ respectively.

The average intensity may be the most suggestive type of information, representing the DC component of the grid cell. However, it is insufficient to describe the intensity distribution. To obtain a better model of the intensity structure, the average intensity gradient is used to capture the intensity trend. In our implementation, we firstly calculate the mean intensity $\bar{I}_i$ and the mean intensity gradient $\bar{I}'_i$ for each grid cell. Then, inspired by the BRIEF descriptor, the average intensity gradient and curvature between any two grid cells i and j in the same angular bin k are encoded by a binary tuple

$$r_{l,k} = \left[H\left(\bar{I}_{i,k} - \bar{I}_{j,k}\right), H\left(\bar{I}'_{i,k} - \bar{I}'_{j,k}\right)\right]^T, \tag{2}$$

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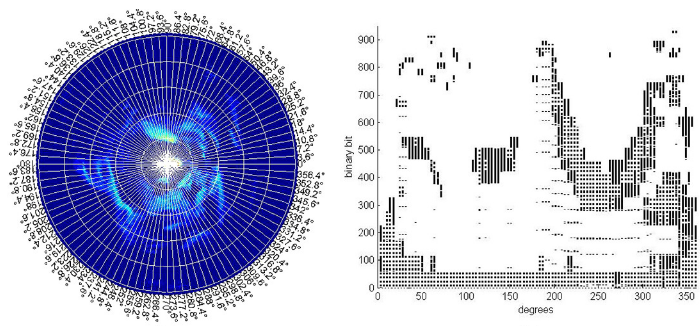

(a) (b)

FIGURE 2. Extraction of rotation feature from the MSIS sonar image. (a) Divide the sonar image into grid cells in polar coordinate. (b) Encode the intensity distribution by the polar gradient matrix, where black represents "1" and white represents "0".

where $H(\cdot)$ is the Heaviside function, $i, j = 1, ..., N_d, i \neq j, l = 1, ..., C_{N_d}^2, k = 1, ..., N_a$, and T denotes transpose. Concatenating all binary tuples in the same angular bin k into a column, we obtain a rotation feature matrix $R = \{r_{l,k}\}$ with the dimension of $2C_{N_d}^2 \times N_a$.

As an example, the polar gradient matrix of the sonar scan in Fig. 2(a) is displayed in Fig. 2(b). The angular resolution of the MSIS sonar is $1.8°$ and the imaging distance is 50 meters. We have empirically set the unit angle to $d\theta = 3.6°$ and the unit distance to $dD = 1.6m$, then the rotation feature is a 930×100 matrix.

2) Rotational Parameter Estimation

Once the binary rotation feature matrices, R_1, R_2, have been extracted from the reference and floating sonar scans respectively, the rotational parameter could be pursued by minimizing the differences between them. That is,

$$\Delta\theta^* = \underset{\Delta\theta}{\arg\min}\, J_\theta \left(R_1, R_2, \Delta\theta\right), \tag{3}$$

where the objective function is

$$J_\theta \left(R_1, R_2, \Delta\theta\right) = \sum_{k=1}^{N_a} XOR \left(R_1\left(k\right), R_2\left(k + \Delta\theta\right)\right), \tag{4}$$

and $XOR(\cdot)$ is the exclusive-OR operation. Note that in order to alleviate the boundary effects, the correlation is computed in a circular way. An example of rotational parameter estimation by correlating the rotation feature matrixes is given in Fig. 3.

B. INTENSITY PROJECTION HISTOGRAM AND TRANSLATIONAL PARAMETER ESTIMATION

In this section, we design a feature descriptor to model the energy distribution along the two orthogonal axes, named intensity projection histogram, which will be further used to estimate the translational parameters.

1) Intensity Projection Histogram

For the translational parameters estimation methods that are based on the cross-correlation, it is better for the intensity distribution function or intensity histogram to have sharp peaks. Therefore, we have to project the pixels in a specifically selected direction. The unit projection vector that passes through the origin and has an angle of θ can be written as $\boldsymbol{v}_\theta = [\cos\theta, \sin\theta]^T$. Then, the offset of a point $\boldsymbol{p} = [x_i, y_i]^T$ on the projection vector is the inner product $d_i = \langle \boldsymbol{p}, \boldsymbol{v}_\theta \rangle = x_i \cos\theta + y_i \sin\theta$. If we discretize the projection range into a set of equal length bins and sum the intensity of all pixels that are projected into the same bin, an intensity projection histogram will be resulted. A similar intensity projection histogram could be obtained in the orthogonal direction of $\theta^\perp$.

Denote the angles of two projection lines in the reference scan as θ_g and $\theta_g^\perp$, then their counterparts in the floating scan can be written as $\theta_f = \theta_g - \Delta\theta^*$ and $\theta_f^\perp = \theta_g^\perp - \Delta\theta^*$. Obviously, the angle of the first projection vector θ_g determines the feasibility of all the four intensity projection histograms.

Here, a heuristic strategy based on entropy minimization is used to determine the angle θ_g of the projection line. We again divide the sonar image into sectors in polar coordinate, now with a larger angular bin size, e.g., $18°$. The echoic intensity distribution in each sector is characterized by the entropy of the intensity histogram. Generally, a concentrated histogram has smaller entropy, which benefits the discrimination between similar scans. Therefore, we choose the starting angle of the sector that has the minimum entropy as θ_g.

2) Translational Parameter Estimation

Once the intensity projection histograms, H_{θ_f}, $H_{\theta_f^\perp}$, H_{θ_g}, $H_{\theta_g^\perp}$, have been extracted from two sonar scans respectively, the translational parameters could be determined by locating the global maximum in the correlation functions. That is,

$$\Delta x_\theta^* = \underset{\Delta x}{argmax}\, J_{trans} \left(H_{\theta_g}, H_{\theta_f}, \Delta x\right), \tag{5}$$

and

$$\Delta y_\theta^* = \underset{\Delta y}{argmax}\, J_{trans} \left(H_{\theta_g^\perp}, H_{\theta_f^\perp}, \Delta y\right), \tag{6}$$

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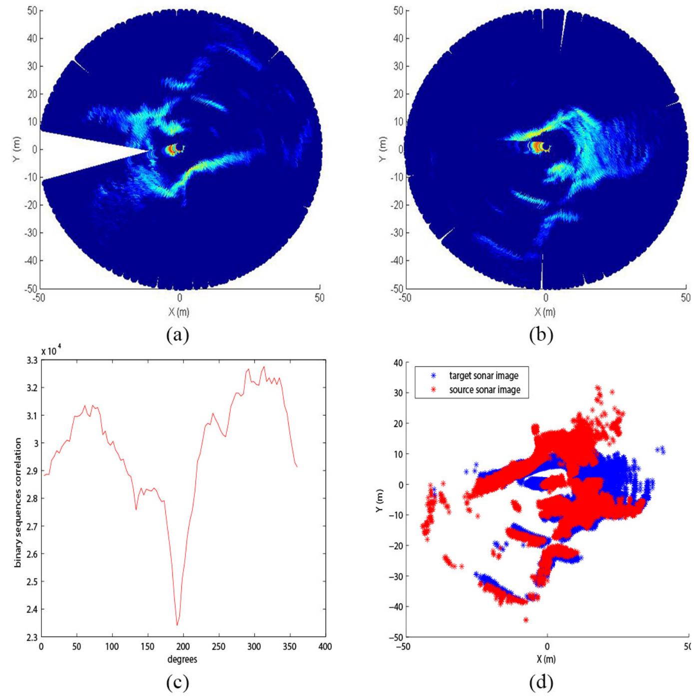

FIGURE 3. An example of rotational parameter estimation by correlating the rotation feature matrixes. Two sonar scans are shown in (a) and (b) respectively, the correlation function between their rotation feature matrixes is displayed in (c). It can be seen in (d) that the two scans are well registered.

where the objective function is

$$J_{trans}\left(H_1, H_2, n\right) = \sum_{i=-N_h/2}^{N_h/2} H_1\left(i\right) \times H_2\left(i-n\right). \quad (7)$$

H_1 and H_2 are the intensity projection histogram of the reference scan and floating scan respectively, and N_h is the histogram length. Note that the results of (3) and (5) to (6) have been multiplied by the angular and projection histogram bins, respectively.

The translation Δx_θ^*, Δy_θ^* could be easily solved by localizing the global maximum in the correlation functions along two orthogonal axes. Then, we have to map it to the physical coordinates by the trivial rotation transformation.

An example of translational parameter estimation by correlating the intensity projection histogram is given in Fig. 4. Although two sonar images are not strictly aligned in Fig. 4(e), the accuracy is not only sufficient for the similarity score calculation, but also beneficial for the further precise matching with the ICP (iterative closest point) algorithm [29] for the loop-closure constraint refinement.

Now, we have explicitly described the methods to coarsely register two sonar scans, and then the similarity matrix could be easily obtained by calculating the registration error (see (1)). Next section will be devoted to the loop-closure detection by filtering the similarity matrix with the PHD filter.

V. LOOP-CLOSURE DETECTION

When the vehicle travels to a place that has been visited before, the correlation within sonar scan sequence between the two visits will introduce a track in the similarity matrix. Besides, due to the long scanning range of MSIS sonar, multiple tracks may simultaneously appear. Loop-closure detection is to extract those tracks from the similarity matrix. However, the accidental correlations between the low-resolution acoustic scans will introduce strong background noise, aggravating the difficulties in track extraction.

In this paper, we analogously take the tracks in the similarity matrix as the tracks of multiple virtual targets. Then, the track extraction becomes a multi-target tracking problem. With respect to the fact that the appearance of tracks is somewhat a random process, we use the PHD filter to detect and track the virtual targets whose number is variable.

It has been demonstrated in Ref. [11] that PHD filter is able to determine the states of multi-targets along with the unknown targets number. Because no data association is required to complete the update step, the PHD filter provides an advantage in scenarios of significant noise clutters and missed detections, especially for similarity matrix constructed from the natural underwater environment. Below, we describe the PHD filter and its implementation under the simplifying assumption of linear Gaussianity.

A. PHD FILTER

Generally, multi-target tracking methods that are based on Bayesian filtering propagate the multi-target posterior density recursively, although with a practically intractable computational cost. With the introduction of random finite sets (RFS), which treat the individual targets and observations

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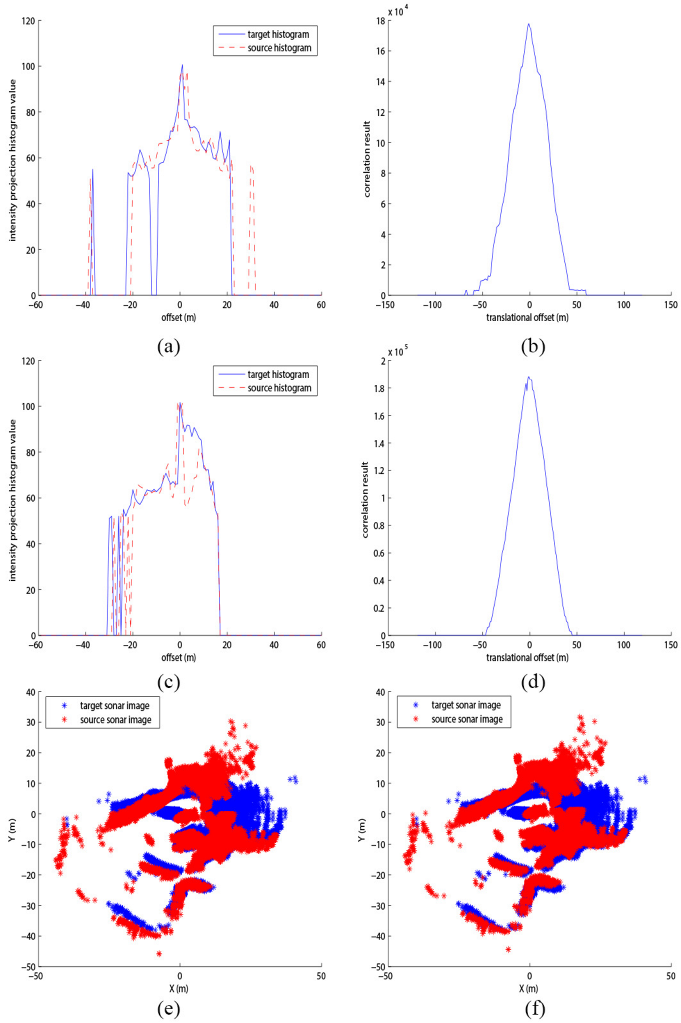

FIGURE 4. The intensity projection histograms along two orthogonal axes are plotted in (a) and (c) respectively, and their corresponding correlation functions are in (b) and (d). The registration result after rotational and translational correction is displayed in (e), while that only with rotational compensation is shown in (f) for comparison.

as two set-valued collections, the PHD filter propagates the first order statistical moment, strongly reducing the computational cost.

The PHD filter consists of the prediction step and the update step [11], which are, respectively, given by

$$\begin{aligned} v_{k|k-1}(x) = \int p_{S,k}(\zeta) f_{k|k-1}(x|\zeta) v_{k-1}(\zeta) d\zeta + \\ \int \beta_{k|k-1}(x|\zeta) v_{k-1}(\zeta) d\zeta + \gamma_k(x) \end{aligned} \tag{8}$$

and

$$\begin{aligned} v_k(x) = \left[1 - p_{D,k}(x)\right] v_{k|k-1}(x) + \\ \sum_{z \in Z_k} \frac{p_{D,k}(x) g_k(z|x) v_{k|k-1}(x)}{\kappa_k(z) + \int p_{D,k}(\xi) g_k(z|\xi) v_{k|k-1}(\xi) d\xi}. \end{aligned} \tag{9}$$

In the prediction step, x is the target state, $v_{k|k-1}(\cdot)$ is the intensity of predicted multi-target states RFS, $p_{S,k}(\cdot)$ is the probability of target survival, $f_{k|k-1}(\cdot|\cdot)$ is the single target motion distribution, $v_{k-1}(\cdot)$ denotes intensity of updated multi-target states RFS at time $k-1$, $\beta_{k|k-1}(\cdot|\cdot)$ is the intensity of spawned targets RFS, and $\gamma_k(\cdot)$ is the intensity of spontaneous birth target states RFS at time k. In the update step, $p_{D,k}(\cdot)$ is the probability of target detection, $g_k(\cdot|\cdot)$ is the single target observation likelihood function, and $\kappa_k(\cdot)$ is the intensity of clutter RFS at time k.

B. GM-PHD FILTER

It has been demonstrated in Ref. [30] that under the assumption that the probability density functions in PHD filter assume a Gaussian Mixture distribution (GM-PHD), it is possible to derive a closed form solution for the prediction and update steps. The Gaussian assumption applies to the dynamical model $f_{k|k-1}(x|\zeta)$, the measurement model $g_k(z|x)$, and the spontaneous birth intensity $\gamma_k(x)$.

The dynamical model of each target and the measurement model are assumed to be linear Gaussian models respectively, i.e.,

$$f_{k|k-1}(x|\zeta) = \mathcal{N}\left(x; F_{k-1}\zeta, Q_{k-1}\right) \tag{10}$$

$$g_k(z|x) = \mathcal{N}\left(z; H_k x, R_k\right), \tag{11}$$

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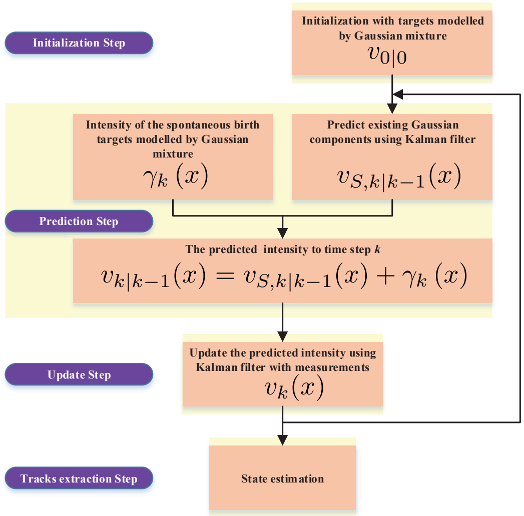

FIGURE 5. Flowchart of GM-PHD filter.

where $\mathcal{N}(\cdot)$ denotes a Gaussian density. F_{k-1} and H_k are the state transition matrix and the observation matrix respectively, Q_{k-1} and R_k stand for the covariances of process noise and observation noise.

With the Gaussian mixture assumption, the intensity of spontaneous birth RFS is

$$\gamma_k(x) = \sum_{i=1}^{J_{\gamma,k}} w_{\gamma,k}^{(i)} \mathcal{N}\left(x; m_{\gamma,k}^{(i)}, P_{\gamma,k}^{(i)}\right), \tag{12}$$

where $J_{\gamma,k}$ is the number of Gaussian components, $w_{\gamma,k}^{(i)}$ is the weight of i^{th} component at time k, $m_{\gamma,k}^{(i)}$ and $P_{\gamma,k}^{(i)}$ are the corresponding mean and covariance respectively.

The survival and detection probabilities are state independent, i.e.,

$$p_{S,k}(x) \equiv p_{S,k} \tag{13}$$

$$p_{D,k}(x) \equiv p_{D,k} \tag{14}$$

and both are empirically set to be 0.9 in the latter experiments.

With the above assumptions, the prediction step and update step in PHD filter, i.e., (8) and (9), could be solved analytically. See Vo *et al.* [30] for more details.

C. LOOP-CLOSING TRACK DETECTION

The state vector of GM-PHD filter is composed of the position and velocity of the virtual target, $x_k = \begin{pmatrix} p_{x,k} & \dot{p}_{x,k} & p_{y,k} & \dot{p}_{y,k} \end{pmatrix}^T$, where $(p_{x,k}, p_{y,k})$ denotes the position and $(\dot{p}_{x,k}, \dot{p}_{y,k})$ the velocity. When a place is revisited in a loop-closing way, it will be often scanned by the sonar in reverse order. Then, the track generated by scan data correlation will be perpendicular to the main diagonal in the similarity matrix. Therefore, we set $\dot{p}_{x,k} = 1$ (step/frame) and $\dot{p}_{y,k} = -1$ (step/frame)

The flowchart of the applications of the GM-PHD filter to loop-closing track detection is displayed in Fig. 5. It is mainly composed of initialization step, prediction step, update step, and track extraction step.

a: Initialization step:

The GM-PHD filter is initialized with a Gaussian mixture,

$$v_{0|0}(x) = \sum\nolimits_{i=1}^{J_0} w_0^{(i)} \mathcal{N}\left(x; m_0^{(i)}, P_0^{(i)}\right). \tag{15}$$

As the similarity matrix M is a lower triangular matrix, it has only one element m_{11} in the first column. Therefore, both the number J_0 and the weight w_0^1 are set to 1.

b: Prediction step:

The predicted intensity for time k is a Gaussian mixture, given by

$$v_{k|k-1}(x) = v_{S,k|k-1}(x) + \gamma_k(x), \tag{16}$$

where $v_{S,k|k-1}(\cdot)$ is obtained by predicting the existing Gaussian components at time $k-1$. In $\gamma_k(\cdot)$, the number of Gaussian components $J_{\gamma,k}$ is set to be 5. It means that, when a new sonar scan is collected, the coordinates of five best-matched scans are used to initialize the positions of newborn targets.

c: Update step:

With the measurements $Z_k = \left\{z_{k,1}, z_{k,2}, ..., z_{k,|Z_k|}\right\}$ at time k, the posterior intensity is updated as

$$v_k(x) = (1 - p_{D,k})\, v_{k|k-1}(x) + \sum_{z \in Z_k} v_{D,k}(x; z), \tag{17}$$

where $v_{D,k}(x; z) = \sum\limits_{j=1}^{J_{k|k-1}} w_k^{(j)}(z) \mathcal{N}\left(x; m_{k|k}^{(j)}(z), P_{k|k}^{(j)}\right)$.

d: Track extraction step:

To extract the tracks of virtual targets, we assign an identity to each Gaussian component [31]. New identities are assigned to the Gaussian terms contributed by the initialization process or spontaneous birth process, and each term becomes the root of a tree whose branch number grows linearly with sequential measurements. More precisely, when measurements Z_k are received at time k, each term of the mixture representing the predicted intensity $v_{S,k|k-1}(\cdot)$ is updated with $|Z_k|$ measurements, forming $1 + |Z_k|$ (considering the missed detection scenario) updated Gaussian terms to which the same identities are assigned as to their prior. Thus, each tree is identified by its unique label that stems from the root component. If none of the weights of the updated Gaussian terms is larger than a predetermined threshold, the tree will not be updated further, i.e., the track of corresponding virtual target terminates. Finally, the track of each virtual target can be extracted by retroactively and iteratively finding the parents of the Gaussian component with the maximum weight in the terminated layer.

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Remark 1 When loop closure occurs, the vehicle revisits the mapped area along a similar path, in the same or opposite direction. On the other hand, it is necessary for the vehicle to cruise with a low velocity when it observes the underwater environment with an MSIS sonar, because the longer scanning period will introduce severe distortions to the scans at high speed. Then, constant velocity could be assumed. Therefore, the linear Gaussian dynamics model, together with the additive Gaussian process noise are feasible for describing the motion of each virtual target. The state transition matrix and the process noise covariance matrix are

$$F_k = \begin{pmatrix} 1 & T & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & T \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad Q_k = \begin{bmatrix} \frac{T^4}{4} & \frac{T^3}{2} & 0 & 0 \\ \frac{T^3}{2} & T^2 & 0 & 0 \\ 0 & 0 & \frac{T^4}{4} & \frac{T^3}{2} \\ 0 & 0 & \frac{T^3}{2} & T^2 \end{bmatrix} \tag{18}$$

respectively, where T is set to be 1 (step/frame) and Q_k is similar to that in [15].

Remark 2 Similarly, with the linear Gaussian assumption, the observation matrix and noise covariance matrix of the observation model are respectively given by

$$H_k = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad R_k = \sigma_m^2 \begin{bmatrix} 1 & 0 \\ 0 & \left(\frac{1}{10}\right)^2 \end{bmatrix} \tag{19}$$

The variance of measurement noise along the y coordinate is set to be smaller than that of x coordinate, because the uncertainty mainly arises from a few similar images near the loop-closing image pair.

Remark 3 In the iteration process, pruning and merging strategies are employed to keep the number of Gaussian components at a tractable level. On the one hand, components with weight $w_k^{(i)} \leq 10^{-7}$ are removed, and two components with Mahalanobis distance less than $U = 0.8$ are merged. On the other hand, only the $J_{\max} = 200$ components with the highest weights are preserved if there are still too many components.

D. LOOP-CLOSURE CANDIDATES VALIDATION

After loop-closing tracks are extracted from the similarity matrix, three heuristic strategies are designed to remove inappropriate loop-closure candidates.

Firstly, if an MSIS sonar scan contains a large gap, i.e., larger than 90 degree, after the correction of motion-induced distortions, the related constraint candidates are discarded, because the missing of a big area of point clouds is likely to fail the ICP algorithm when pursuing the precise transformation parameters.

Secondly, a loop-closure candidate will be removed if the registration error is unable to drop below an empirically determined threshold in a given time in the precise registration stage using ICP algorithm.

Finally, although there is drift error in the dead-reckoning process, the spatiotemporal continuity contained in the topology can be a useful measure for false positive loop-closure constraints detection [32]. In the latter experiments, the displacement difference between loop-closing poses before and after pose graph optimization, together with the information matrix, are used to filter the loop-closing constraints.

Loop-closure detection is often evaluated by the precision and recall ratio. However, it is not suitable for the MSIS images because of the longer imaging distance and the lower resolution. To evaluate the performance of the proposed loop-closure detection pipeline, we incorporate the validated loop-closure constraints into the GraphSLAM [33] framework to optimize the vehicle trajectory and construct the global map. The optimized trajectory can be compared to the GPS trajectory when available, and the consistency of global map can be judged by the consistency of loop-closing places. In the next section, we will introduce the global error adjustment based on the GraphSLAM framework.

VI. GLOBAL ERROR ADJUSTMENT

The global error adjustment is implemented in the GraphSLAM framework, which poses the SLAM problem as a pose graph optimization problem. A graph is made up of nodes connected by edges. Let $\boldsymbol{L} = \{\boldsymbol{l}_i\}_{i=1,\ldots,N}$ be a collection of nodes, where node $\boldsymbol{l}_i$ contains the robot pose and sonar scan. Usually, the pose of underwater vehicle is provided by the inertial navigation system. The edge connecting node $\boldsymbol{l}_i$ and node $\boldsymbol{l}_j$ is characterized by the error $\boldsymbol{e}_{ij}(\boldsymbol{l}_i, \boldsymbol{l}_j)$ and the information matrix $\boldsymbol{\Omega}_{ij}$.

The error

$$\boldsymbol{e}_{ij}(\boldsymbol{l}_i, \boldsymbol{l}_j) = \boldsymbol{z}_{ij}(\boldsymbol{l}_i, \boldsymbol{l}_j) - \hat{\boldsymbol{z}}_{ij}(\boldsymbol{l}_i, \boldsymbol{l}_j) \tag{20}$$

records the transformation parameter difference between the measurement $\boldsymbol{z}_{ij}(\boldsymbol{l}_i, \boldsymbol{l}_j)$, which is obtained by finely registering sonar scans by ICP algorithm, and the estimate $\hat{\boldsymbol{z}}_{ij}(\boldsymbol{l}_i, \boldsymbol{l}_j)$, which is provided by the dead-reckoning system. In the latter discussion, $\boldsymbol{e}_{ij}(\boldsymbol{l}_i, \boldsymbol{l}_j)$ is short for $\boldsymbol{e}_{ij}$. The measurement is modeled by a Gaussian function, $\boldsymbol{z}_{ij} \sim \mathcal{N}\left(\boldsymbol{z}; \boldsymbol{\mu}_{ij}, \boldsymbol{\Omega}_{ij}^{-1}\right)$, with the mean $\boldsymbol{\mu}_{ij}$ provided by sonar scan registration and covariance given by the Hessian matrix method [34].

The measurement, that is used as a constraint by us, is the basis for global error adjustment. Let C be the constraint set. The goal of the GraphSLAM is to find the configuration of the nodes $\boldsymbol{L}^*$ that minimizes the objective function given as follows

$$\boldsymbol{L}^* = \arg\min_{\boldsymbol{L}} F(\boldsymbol{L}) = \sum_{\langle i,j\rangle \in C} \boldsymbol{e}_{ij}^T \boldsymbol{\Omega}_{ij} \boldsymbol{e}_{ij} \tag{21}$$

Once the optimized vehicle trajectory $\boldsymbol{L}^*$ is obtained, a consistent map can be accordingly generated by stitching the sonar scans. We refer the reader to [33] for more details about the GraphSLAM framework.

Note that edges exist both between consecutive nodes and nonconsecutive nodes. However, only the edges between nonconsecutive nodes are considered, because the dead-reckoning error between consecutive nodes is negligible for efficient global error adjustment. When a sharp change

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between consecutive poses occurs, a new constraint should be added to the constraint set, because the dead-reckoning estimation error is so large it will affect the global error adjustment.

VII. EXPERIMENTAL RESULTS

In this section, we use two datasets to study the performance of the proposed loop-closure detection method. The first dataset, provided by Ribas *et al.* [2], was collected using a Tritech Miniking sonar when the AUV traversed through the Fluvia Nautic abandoned marina near St. Pere Pescador on the Costa Brava. The second dataset, provided by Mallios *et al.* [35], was acquired in a natural underwater cave located in the L'Escala area of Costa Brava, Spain. The difference between the two is obvious, as the marina is an artificial, while the cave is a natural environment. Later, we will refer to them as the marina dataset and the cave dataset, respectively. It is worth to note that the marina dataset has GPS data, so we can evaluate the effectiveness of the proposed method objectively.

The marina is a fully structured man-made environment while the cave is a fully unstructured natural environment. This difference in the environments has a significant influence on the modules of the proposed loop-closure detection pipeline. For example, the intensity projection histogram generated by the cave dataset is more discontinuous than that of the marine dataset, adding a large number of random noise into the similarity matrix, due to the wide scene correlation in the natural environment. Further, the background noise challenges the PHD-based loop-closing tracks extraction process, with more false positive loop-closing tracks aggravating the difficulty in loop-closure candidates validation.

A. THE MARINA DATASET

A satellite map of the marina is shown in Fig. 6(a). The dataset includes a small loop around the principal water tank (approximately 400m) and a 200m straight path through an outgoing canal. The MSIS sonar scans the whole 360 degree sector at a range of 50m with 0.1m resolution and 1.8 degree angular step. A differential global positioning system (DGPS) unit mounted in a buoy was attached at the top of the vehicle during the experiment to provide the ground truth motion trajectory with the theoretical drift error no more than 1.22m.

The similarity matrix constructed from the marine dataset by registering the sonar scans with the polar gradient matrix and the intensity projection histogram is shown in Fig. 6(b). Note that the similarity scores smaller than a given threshold of ε are shown as black dots in Fig. 6(c). The first 13 frames are ignored because navigation data are unavailable. In practical implementations, we focus on identifying large loops, where substantial accumulated errors need to be corrected. Therefore, we remove also the ten off-diagonals next to the diagonal in the similarity matrix, which means that we ignore sonar scans that are known to be collected at a very short distance. The loop-closing tracks extracted by the GM-PHD filter are displayed in Fig. 6(d). Since the dataset contains GPS data of the vehicle's motion trajectory, the GPS coordinates of each sonar image can be known. Therefore, it can be judged whether the two sonar images are collected from the same place. A simple comparison to the GPS data shows that the labeled tracks displayed in Fig. 6(d) are composed of genuine loop-closing scan pairs. On the other hand, all the black dots except those near the main diagonal are close to the loop-closing tracks. It demonstrates that the proposed sonar scan similarity measure, including the

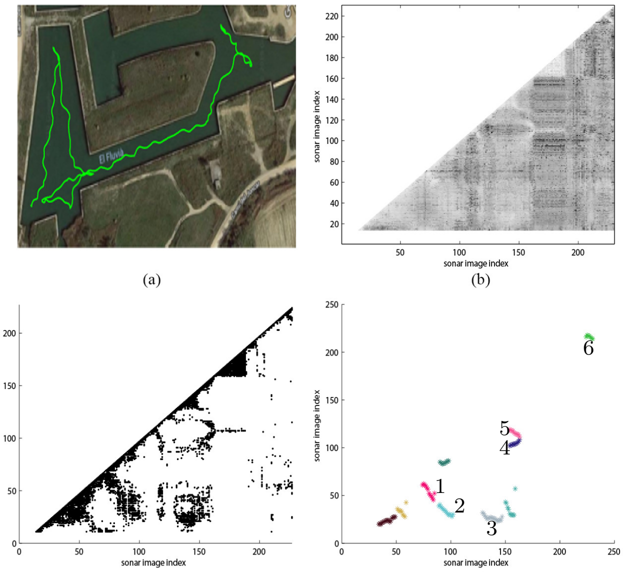


FIGURE 6. Loop-closure detection for the marine dataset. (a) Bird's eye view (Google) of the marina dataset collection site with the motion trajectory of the vehicle. (b) Grayscale image of similarity matrix constructed from marina environment. (c) displays the similarity matrix after threshold and (d) presents the loop-closing tracks extracted by GM-PHD filter.

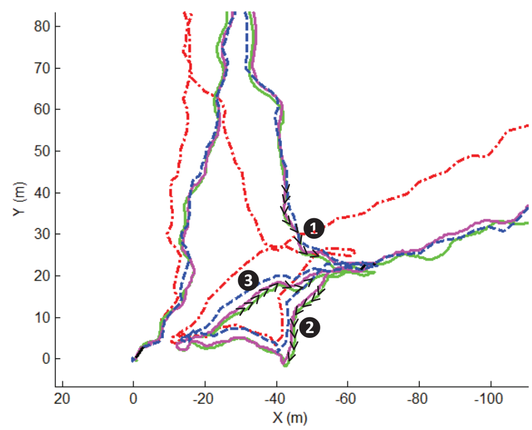
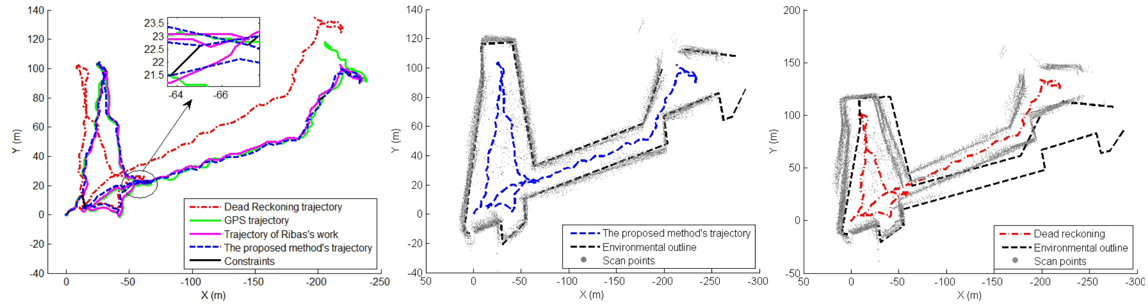


FIGURE 7. The vehicle motion direction is indicated by the black arrows. Due to the long scanning range of the MSIS sonar, the images generated during the vehicle traversing path 3 strongly correlated with the counterparts of path 1 and 2, simultaneously forming track 4 and 5.



(a) (b) (c)

FIGURE 8. Motion trajectories and environment map after global error adjustment. (a) Trajectories generated by different methods. Note that the loop-closure constraints are plotted in short solid black line. The panoramic map generated by stitching the sonar scans along our proposed method's motion trajectory is plotted in (b), while that along the dead-reckoning trajectory is displayed in (c) for comparison. The real physical world is represented as black dash line.

polar gradient matrix and the intensity projection histogram, is able to efficiently measure the similarities and differences between low-resolution MSIS sonar scans.

Each track refers to a potential loop closure.

- Tracks 1, 2 are formed when the vehicle reversely revisited the area along the path $x = -45$ (see Fig. 8(a) downwards).
- The third loop-closure point, near (-10,4), is close to the starting position. It can be seen that track 3 is nearly parallel to the x axis, because the vehicle stayed there for about three minutes.
- It is interesting to see that two tracks are simultaneously formed in the same time interval 2128s-2266s (corresponds to the sonar image index of 154-164 in Fig. 6(d)). The underlying reason is that it is the third time for the vehicle to pass the place around (-50,20) with the long scanning range MSIS, so current scan sequence is strongly correlated with that in the former two visits, as shown in Fig. 7 which is a magnified part of Fig. 8(a). It demonstrates that the GM-PHD filter is able to extract the variable number of loop-closing tracks.
- Track 6 corresponds to a small loop closure located near (-230,100).

In Fig. 8, we show the final motion trajectory of the vehicle after global error adjustment, and the environment map generated by stitching the sonar scans along the estimated motion trajectory. In Fig. 8(a), the trajectories provided by dead reckoning, GPS signal, and the proposed method are plotted in dash-dot (red), solid (green), and dash (blue) respectively. The trajectory generated by the Ribas *et al.* [2] is plotted in magenta solid line for comparison, and the loop-closure constraints are plotted in short solid black line for understanding. One can see that dead-reckoning trajectory suffers from a noticeable drift, however, the SLAM trajectory of our method follows the GPS track with considerable precision benefiting from the loop-closure detection.

In order to analyze the effects that the loop-closure constraint has in the full SLAM system, a panoramic map is generated by stitching the sonar scans along the estimated motion trajectory. The maps and associated SLAM and dead-reckoning trajectories are plotted in Figs. 8(b) and 8(c), respectively. A simple measure called "crispness" is adopted to quantitatively measure the consistency of the generated global map. Explicitly, we divided the global map into equally sized voxels and counted the occupied voxels, with a lower value indicating higher consistency. We refer the reader to [36], [37] for more detail. The map generated by the proposed filtering method has sharper walls (its crispness measure is 11359), which is also consistent with the physical environment. However, the map generated by the dead reckoning has obvious ghosting phenomena, which leads to a crispness of 12159. It not only demonstrates that the proposed PHD filtering of the similarity matrix is capable of detecting the loop-closing constraints, but also shows that the estimation of the transformation parameters obtained by feature correlation provides a feasible initialization for the ICP algorithm and thus for the final loop-closure constraint refinement.

To further quantitatively evaluate the performance of the proposed loop-closure detection method, we list the location estimation error between the trajectory generated by each method and the GPS ground truth in Table 1. The error statistic data for MSISpIC method [38], Feature-based SLAM [2] and Pose-based SLAM [19] are taken from [19] for comparison. The available methods that have been tested in the marine dataset can be divided into two categories, with the first one using dead reckoning and scan registration (refer to "DR+SR" in Table 1) and the second one additionally including loop-closure detection (refer to "DR+SR+LCD" in Table 1). To objectively study the performance of our proposed loop-closure detection method, scan registration

TABLE 1. Location estimation errors of different methods

Method/error	*Modules*	**Mean**	**SD**	**Min**	**Max**
DR	*DR*	17.46	11.47	0.09	45.42
MSISpIC	*DR+SR*	3.99	2.11	0.03	8.49
Feature-based SLAM	*DR+SR+LCD*	1.91	1.32	**0.01**	5.08
Pose-based SLAM	*DR+SR+LCD*	1.90	1.09	**0.01**	**4.93**
Our method	*DR +LCD*	**1.58**	**0.98**	0.09	5.85

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between consecutive scans is excluded in our SLAM system and only the loop-closure detection module is adopted to reduce the global accumulated error.

Three conclusions can be drawn from the location estimation error statistics in Table 1. Firstly, all the listed methods that include scan registration or loop-closure detection modules can drastically reduce the location estimation error included in the dead-reckoning process. Secondly, the performance of Feature-based SLAM and Pose-based SLAM is superior to MSISpIC method. Loop-closure detection such as feature-matching and point cloud registration, or data association such as scan matching alone can effectively reduce the accumulated error. Nevertheless, we show that the employment of both two strategies can further improve the location estimation precision. Because, a potential risk that the drift error will grow rapidly if the scan matching error is accidentally large, is very likely to happen when there is a rotation of U-turn. Lastly, the proposed method does not include scan registration between neighboring frames for local error correction, but achieves a comparable performance with Feature-based SLAM and Pose-based SLAM methods, indicating that loop-closure detection may be more important for SLAM systems than the neighboring poses scan registration. The location error is expected to be further reduced if the module that performs scan registration between neighboring scans is incorporated into the proposed pipeline. It is worthy to note that the loop-closure detection module in Feature-based SLAM [2] and Pose-based SLAM [19] depends on the vehicle pose estimation to generate loop-closure candidates, which is unnecessary in our proposed method.

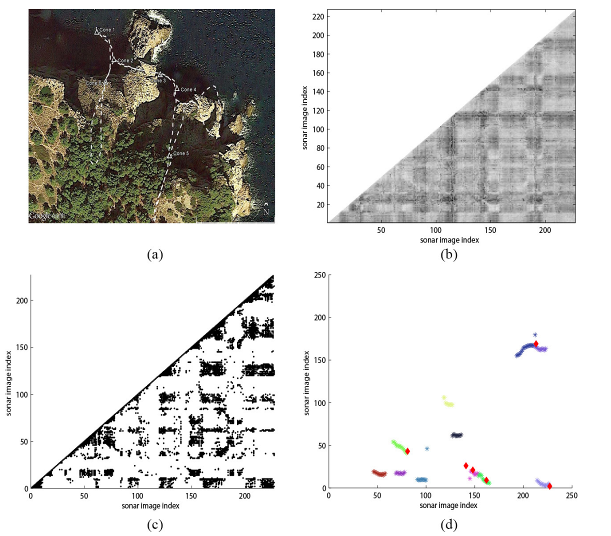

FIGURE 9. Loop-closure detection for the cave dataset. (a) The (Google) map with the motion trajectory of the vehicle and the cone locations. (b) The grayscale image of the similarity matrix constructed from cave dataset. The similarity matrix after threshold is displayed in (c) and the loop-closing tracks extracted by GM-PHD filter are presented in (d).

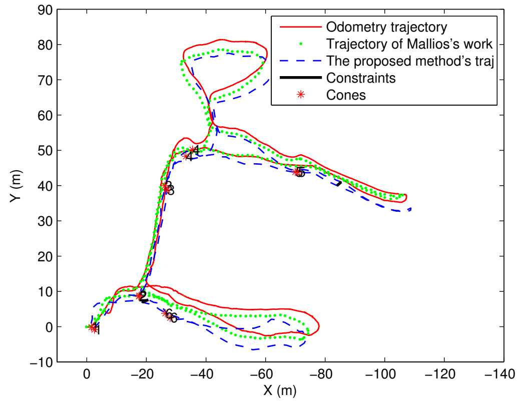

FIGURE 10. The trajectories generated by dead reckoning, Mallios's work [1] and the proposed method are plotted in solid (red), dotted (green), and dashed (blue) lines, respectively. Red asterisks denote traffic cones and loop-closure constraints are shown as black lines near cone 2 and cone 5.

B. THE CAVE DATASET

A map and the overlaid rough motion trajectory of the vehicle and the landmark location of cones are shown in Fig. 9(a). The entire motion trajectory was approximately 500m long. The AUV was equipped with two orthogonal MSIS sonars in order to scan the horizontal and vertical section. The scanning range of the horizontal MSIS was 50m with a 0.126m resolution, and 20m with 0.4m resolution for the vertical one. It is difficult to obtain the ground truth in a complex confined underwater environment like the present one. To validate the algorithms, six traffic cones as ground truth points were strategically placed in locations where the vehicle passed over twice. A low cost analog video camera was mounted looking down to identify the cones.

The similarity matrix constructed from the cave dataset is shown in Fig. 9(b), and the thresholded matrix is displayed in Fig. 9(c). It can be seen that the similarity matrix of the cave dataset contains much more noise than that of the marine dataset (refer to Fig. 6(c)). Because, it is much more difficult to extract discriminative features from the MSIS scans, then the accidental correlations bring in more noise into the similarity matrix.

The loop-closing tracks extracted by the GM-PHD filter are displayed in Fig. 9(d). The six red diamonds correspond to six traffic cones, which represent the known loop-closure constraints. Five diamonds locate on the loop-closing tracks, facilitating the validation of the proposed method that detects loop closures by filtering the noisy similarity matrix with the GM-PHD filter. The loop-closure constraint corresponding to cone 4 (refer to Fig. 9(d)) is not on any loop-closing tracks. Perhaps due to the changes in viewpoints, the image sequences near cone 4 during two visits are extremely different from each other.

Similar to the marine dataset, false positive loop-closure constraints could be extracted during the filtering process. With the heuristic strategies discussed in Section V-D, only

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two loop-closure candidates were maintained in the final constraint set, see the black line segments located near cone 2 and cone 5 in Fig. 10. Although the number of constraints is much less than that in Mallios *et al.* [1], it has been manifested in Ref. [23] that a single, reliable detection of loop closure is all that is required for the map to be corrected.

In Fig. 10, we plotted the trajectories generated by the proposed method (blue dash line), dead reckoning (red line) and Mallios's method (green dot line). For convenience, the final loop-closure constraints were plotted in black lines, and the traffic cones were denoted by red asterisks. It can be seen that, after trajectory optimization by GraphSLAM, both the proposed method and Mallios's method are capable of correcting the accumulated error in dead-reckoning process, especially when the vehicle returns to the starting point at the end of the survey.

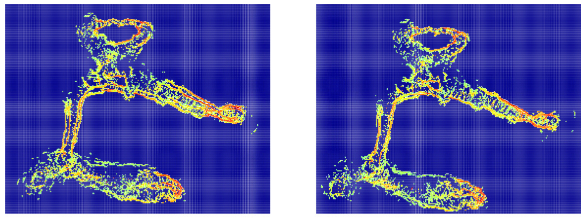
The cone locations were carefully chosen so that the vehicle would pass them twice. Now that the physical locations of the cones were fixed, the difference between two estimates of the same cone could be used to quantitatively measure the accuracy of the estimated vehicle motion trajectory. For clarity, the cone locations estimated by dead reckoning and by Mallios *et al.* [1] are not plotted in Fig. 10. However, the location estimation errors for different methods are displayed in Table 2.

TABLE 2. Location estimation errors for traffic cones by different methods

Cone Num	DR	Mallios's method	Our method
Cone 1	6.6	2.2	**1.22**
Cone 2	3.84	1.68	**0.33**
Cone 3	2.81	**0.28**	1.42
Cone 4	3.53	**0.49**	2.89
Cone 5	2.44	1.4	**0.93**
Cone 6	4.37	2.55	**2.08**

Clearly, loop-closure detection, including the pose threshold method by Mallios *et al.* [1] and GM-PHD filtering in the proposed method significantly reduces the dead-reckoning error. The location estimation errors of four cones are smaller than that of Mallios *et al.* [1], demonstrating that the proposed loop-closure detection pipeline is feasible for the natural underwater environment. The reason for a much higher location error for cone 4 lies in that the corresponding sonar scan pair is not included in the constraint set, due to the absolute difference in sonar image intensity distribution.

In Fig. 11, we show the global maps obtained by our method and dead reckoning. The global map was generated by projecting the scan points into 2D space according to the motion trajectory estimated by the SLAM system or dead-reckoning system. In can be seen in Fig. 11(a) that there are duplicated walls along the top horizontal cave and vertical tunnel, which are resulted from the drift error in the dead-reckoning process. Comparatively, the map after loop-closure detection and global error adjustment is more consistent, and the cave walls are much thinner and sharper. On the other hand, the crispness measurement for Fig. 11(b) is 5926, i.e., smaller than the 6348 for Fig. 11(a). This measurement



(a) (b)

FIGURE 11. The natural underwater cave maps generated by (a) dead reckoning (b) the proposed filtering method.

demonstrates again that loop-closure detection is able to reduce the globally accumulated error.

VIII. CONCLUSION AND FUTURE WORK

In this paper, we have proposed a method to detect loop closure in an underwater acoustic SLAM system that filters the similarity matrix by a PHD filter. Firstly, a similarity matrix is constructed by registering each sonar scan pair and counting the overlaps between them. To register the low resolution sonar scans, the intensity projection histograms and the polar gradient matrix are specially designed to describe the intensity distribution along two perpendicular projection axes and the azimuthal direction. Secondly, to reduce the interference of random noise from accidental correlations, we analogously consider the loop-closing track detection problem as a tracking problem for multiple virtual targets, and successfully extract the loop-closure constraints by a PHD filter. Lastly, the loop-closure constraints are fed into the GraphSLAM system to correct the globally accumulated error and generate the panoramic underwater environmental map. Its application to two MSIS datasets taken from the real underwater environment demonstrates that the proposed translational and rotational feature extraction methods successfully capture the differences between the low-resolution MSIS sonar scans on one hand, and the proposed loop-closure detection method that filters the similarity matrix by PHD filtering is feasible for underwater acoustic loop-closure detection and global error adjustment on the other hand. Its superiority over other methods lies in generating the loop-closure candidates with only sonar scans, independently of vehicle's pose information estimated by dead reckoning.

The current development of energy and control technology enables underwater vehicles to perform long range, large scale and long term missions. When the robot needs to return to the place of departure, it is impossible to precisely determine its location solely by the inertial navigation system, due to the unbounded drift error. In such cases, our proposed loop-closure detection method, which considers sonar scan association, provides preferable choice.

Admittedly, although our method has a good performance in different environments, the construction of the similarity matrix includes a heavy computational cost. With time passing, more and more sonar scans will be collected, so that the dimension of similarity matrix grows rapidly. Each new scan will add a new column to the similarity matrix, because of the necessity of the comparisons of each new scan with all previous ones in order to detect loop closure. More work is required to reduce the complexity of the similarity determination, e.g., by a hierarchical organization of the scans, in order to achieve a higher efficiency in loop-closure detection.

Our future work will focus on the following points: 1) Key-scan detection. The sonar has a long imaging distance, i.e. images collected from adjacent locations are very similar to each other. Thus, a heuristic strategy to limit the dimensionality of the similarity matrix is to detect key frames and calculate the similarity between them. In this way, the computational complexity can be reduced considerably. 2) Filtering of the grayscale similarity matrix instead of the binary matrix. Using a binary matrix can significantly reduce the computational complexity, however with the risk of missing true loop-closure constraints due to thresholding. Therefore, we will modify the PHD filter to accommodate the similarity matrix. 3) Multimodal information fusion. It is widely acknowledged that fusing information from other sensors will improve the environment cognition ability of the underwater vehicle. In the future, we will try to incorporate information from a multibeam echosounder, an optical camera, and a vertical profiling sonar into the underwater SLAM system.

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